

AIoT Landslide & Disaster Early Warning System

Live Autonomous Hotspot Web Dashboard, Rain Simulation & Predictive Analytics

Category: IoT & Disaster Management (Sasthramela)	Microcontroller: ESP32 Dual-Core Xtensa (240MHz)
Sensors: Dual Ultrasonic (HC-SR04), Soil ADC, DHT22	Actuator: Rain Emulation Pump (Relay GPIO 22)
Network: Standalone Wi-Fi Hotspot (192.168.4.1)	Prediction Engine: Multivariate Risk Assessment

1. Executive Summary & Problem Statement

Western Ghats regions in Kerala (such as Wayanad, Idukki, and Pathanamthitta) face severe recurring threats from rainfall-triggered slope failures, debris avalanches, and flash flooding. Traditional meteorological systems provide regional macro-scale warnings that fail to capture hyper-localized slope saturation and runoff dynamics.

This project introduces an autonomous, decentralized **AIoT Environmental Monitoring & Disaster Early Warning System**. Built on the ESP32 platform, it integrates a multi-sensor array measuring volumetric soil moisture, dual acoustic distance channels (stream level & catchment basin), and ambient atmospheric conditions (DHT22). A built-in **Rain Simulation Pump** operated via an optocoupler relay emulates precipitation events in laboratory and science fair demonstrations. The ESP32 hosts its own standalone Wi-Fi Access Point with an embedded real-time web dashboard and an onboard **Multivariate Disaster Prediction Engine** that computes live hazard probabilities and actionable safety alerts.

2. Scientific Principles & Predictive Risk Formulation

A. Soil Moisture Pore-Water Pressure & Landslide Mechanics

According to Terzaghi's effective stress principle ($\sigma' = \sigma - u$), water infiltration increases pore-water pressure u , reducing effective stress and compromising the shear strength of slope soil ($\tau = c' + (\sigma - u) \tan \phi'$). Once saturation exceeds critical thresholds (>75%), slope instability becomes imminent.

B. Multivariate Predictive Hazard Scoring

- The on-device predictive engine evaluates four continuous telemetry dimensions to calculate an instantaneous Hazard Index (0 - 100%):
- **Soil Saturation Factor (45% Weight):** Continuous volumetric water content from ADC GPIO 34.
 - **Precipitation Emulation Status (25% Weight):** Active rain pump relay status (GPIO 22).
 - **Stream Surge / Runoff Level (20% Weight):** Ultrasonic distance tracking rapid catchment accumulation.
 - **Atmospheric Humidity (10% Weight):** Ambient vapor saturation index from DHT22.

3. Hardware Pin Mapping & Circuit Interconnections

Hardware Module	Pin Function	ESP32 Pin	Interface & Operational Logic
Soil Moisture Sensor	Analog Output	GPIO 34	ADC1_CH6 (12-bit Analog: 0-4095)
Ultrasonic Sensor 1 (Stream)	TRIG / ECHO	GPIO 25 / GPIO 26	Stream Surface Distance / Level Tracking
Ultrasonic Sensor 2 (Catchment)	TRIG / ECHO	GPIO 27 / GPIO 32	Reservoir / Catchment Water Depth
Rain Simulation Pump Relay	Relay Control	GPIO 22	Active LOW Optocoupler Relay (LOW=ON)
DHT22 Digital Sensor	Data Pin	GPIO 33	Digital Temperature & Relative Humidity
ESP32 Wi-Fi SoC	SoftAP Engine	Antenna	Standalone Hotspot at 192.168.4.1 (No Pass)

4. Embedded Web Dashboard & Telemetry Architecture

Endpoint / Route	HTTP Method	Payload / Format	Functionality
/	GET	HTML5 / CSS3 / JS	Renders offline dark-theme telemetry dashboard
/data	GET	JSON Object	Real-time stream of temp, hum, soil, u1, u2, relay, hazard
/relay?state=1	GET	JSON State	Energizes relay (GPIO 22 LOW) to activate rain pump
/relay?state=0	GET	JSON State	De-energizes relay (GPIO 22 HIGH) to stop rain pump
Captive Portal	DNS 53	Auto-redirect	Automatically redirects connecting devices to 192.168.4.1

5. Bill of Materials (BOM) & Cost Breakdown

Sl	Component Description	Qty	Unit Cost (INR)	Total (INR)
1	ESP32-WROOM-32 Dual-Core Development Board	1	Rs. 350	Rs. 350
2	HC-SR04 Ultrasonic Distance Sensors	2	Rs. 70	Rs. 140
3	DHT22 High-Precision Digital Temp & Humidity Sensor	1	Rs. 160	Rs. 160
4	Corrosion-Resistant Soil Moisture Sensor Probe	1	Rs. 80	Rs. 80
5	1-Channel 5V Optocoupler Relay Module	1	Rs. 50	Rs. 50
6	12V Submersible Rain Emulation Water Pump	1	Rs. 220	Rs. 220
7	12V Power Supply & Step-Down Buck Converter	1	Rs. 180	Rs. 180
8	Catchment Basin, Model Slope & Wiring Hardware	1 set	Rs. 170	Rs. 170
TOTAL ESTIMATED PROJECT BUDGET				Rs. 1,350 INR

6. Kerala Sasthramela Viva Voce Defense & Key Questions

Q1: How does the system function if internet connectivity is completely lost during extreme weather?

Defense: The ESP32 acts as a fully self-contained Access Point (Hotspot) running an embedded web server and local risk algorithm. No external internet, cloud router, or mobile data is required; emergency personnel can connect directly to 192.168.4.1 to monitor data and control pumps.

Q2: What is the purpose of the web-controlled relay switch?

Defense: The relay operates an emulation pump that sprinkles water over the terrain model, allowing live demonstration of real-time soil saturation progression, runoff accumulation, and dynamic hazard index escalation before the Sasthramela judging panel.